The card

Trunk section: §6.3 The card. The largest domain of the desk.

Talks to: Sources (§6.1) for every external fact; Tools and screeners (§6.2) for candidates; the Market (§6.4) for market context; Research data (§6.5) for the CANSLIM reading; Learning (§6.7) for the score; Registers (§6.6), which freeze it.

Built on: CARD-FIELDS (97 fields), CARD (layout and display), DATA-PIPELINE (how each value is fetched and computed), CANSLIM §7, §11–§14 (the locked CANSLIM card spec and what was built of it), and the code in `src/sideA` – `sideG` and `public/index.html`.

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1. What a card is

A card is **everything the desk knows about one ticker on one trading day**, on one surface the trader reads in seconds, with the depth one tap away.

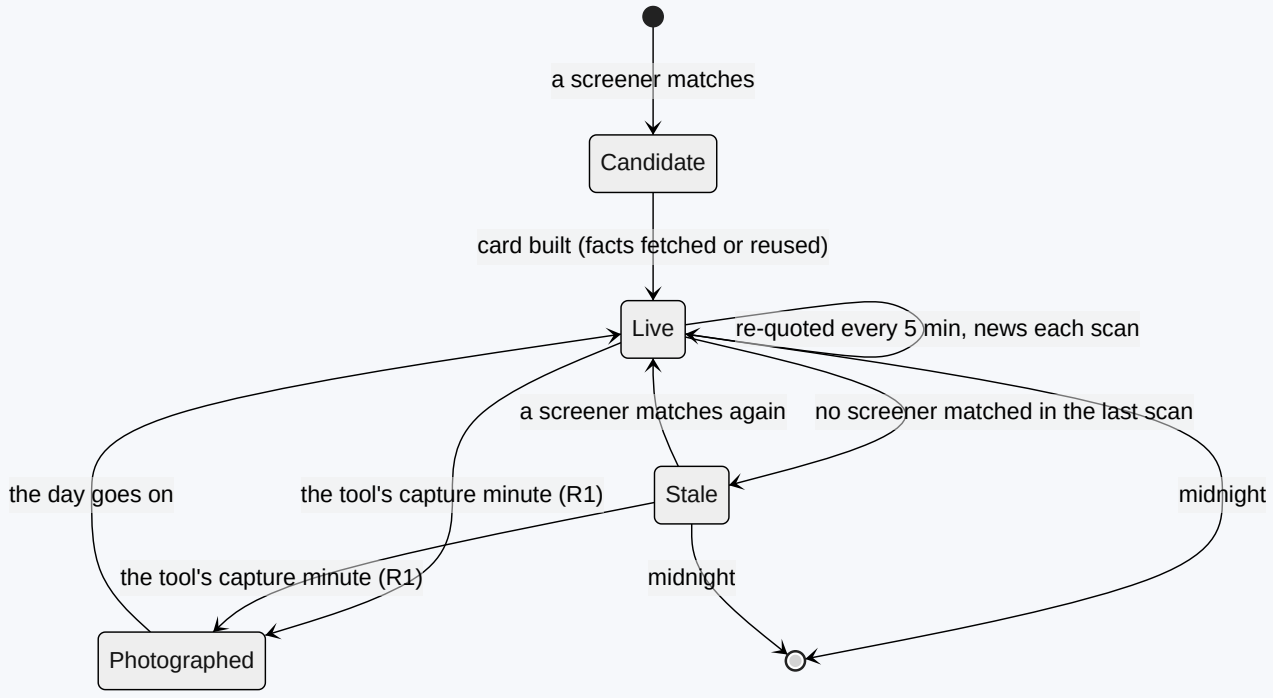
It answers, top to bottom:

1. What is it, and what is it doing right now? (ticker, price, change, live)
2. Why is it here? (which screeners, since when, any catalyst)
3. How does its price sit against its levels? (MAs, VWAP, previous close, open, pre-market high and low, ranges)
4. How much fuel? (relative volume, float, market cap, short interest, borrow)
5. What is the story? (news, catalyst)
6. Is it an O’Neil leader? (the CANSLIM reading)
7. Is the market with it? (regime, trend stages, sector)
8. How likely is it to move, by this tool’s history? (the score)

Two layers. In the old desk each tool built its own copy of every card. In the new design:

Layer	Holds	One per
card	every fact about the ticker: quote, derived numbers, short interest, borrow, news, catalyst, market context, CANSLIM	ticker × trading day (shared by every tool)
tool card	what one tool adds: which of its screeners matched and when, first seen, live or stale, its star, its score, the trader's manual bias	tool × ticker × trading day

Its day



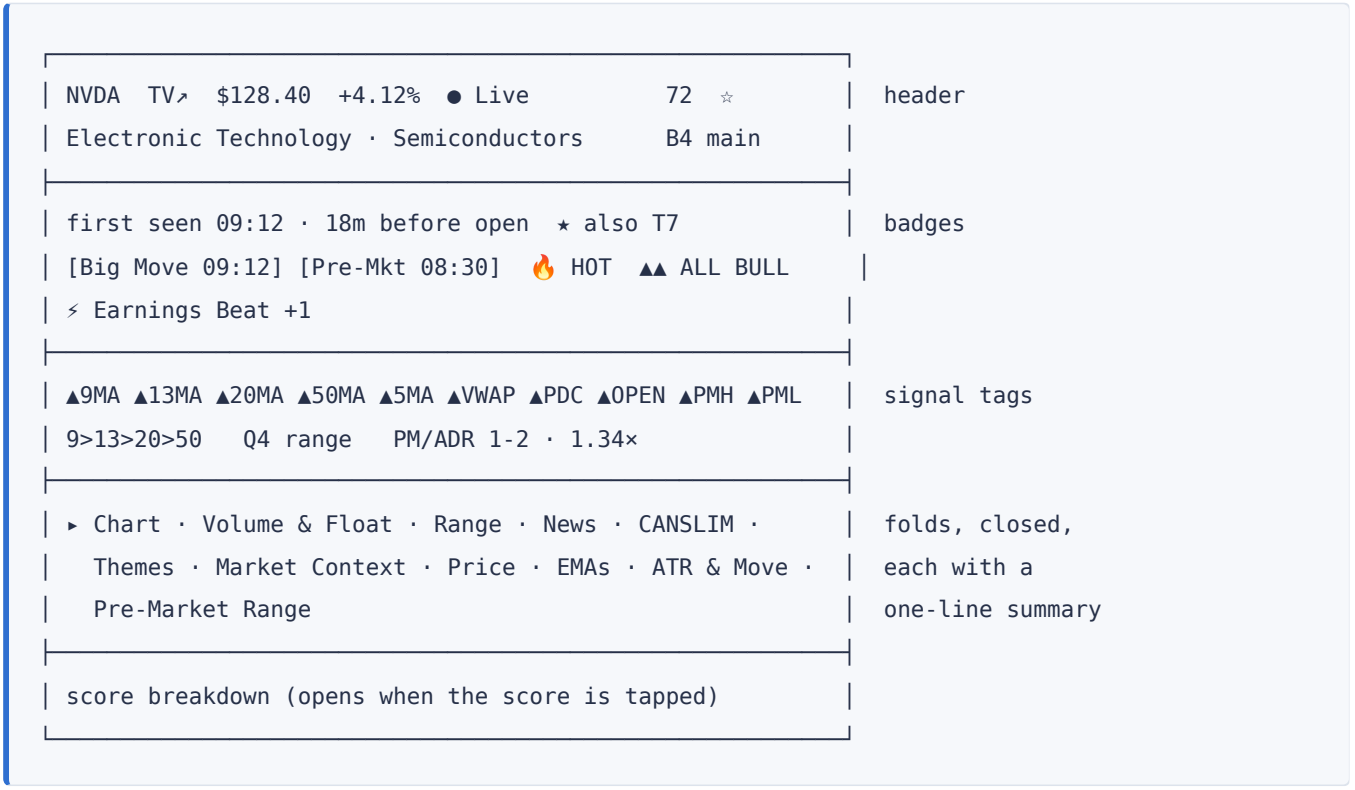
A stale card stays on screen, dimmed, and keeps being re-quoted. The photo takes live and stale cards alike.

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2. The legacy card in full

Every field below is kept in the new design. The `#` column is the field's number in CARD-FIELDS, so each can be traced.

2.1 Layout



One level of folding: every section is a fold, closed by default, and each closed fold shows its key numbers on its heading line.

2.2 Header and badges

#	Field	Source	Rule	Blank means
1	Ticker	TradingView	tap opens the full chart	never blank
—	`TV ↗`	TradingView symbol	links to `tvSymbol` (e.g. `NASDAQ:AAPL`)	—
2	Price	TradingView `close`	plain text: a price is a fact	never blank
3	Change %	TradingView `change`	green up, red down	never blank
4	Live / Stale	desk	Live = matched in the last scan; Stale = matched earlier today; stale is dimmed	outside the live window
5	Sector · Industry	TradingView labels	—	a fault: TradingView always sends them
6	Score	the tool's model	0–100; ≥ 70 high, ≥ 40 mid; tap opens the breakdown	today always `–` (scorer off, D-D1)
7	Table label (`B4 main`, `B4 sub:STRONG_UP`)	the model	which scoring table produced the score; tap opens it	as 6
—	☆ / ★	this tool's shortlist	yellow: the trader's own mark	—
8	First seen `HH:MM` · `Nm before open` / `at the open` / `+Nm`	desk (`seenAt`)	earliest match by any of this tool's screeners; pre-market, early (≤ 30 min after the open) and late styled differently	a fault
—	★ also T2 T7	the unified shortlist	violet: another tool's opinion	not shortlisted elsewhere
9	Screener badges, each with its first-match time	desk	one per screener that matched today; faded when it did not match in the last scan; tooltip gives its window	—
10	🔥 HOT	market engine	the card's sector is hot	not hot
—	▲▲ ALL BULL / ▼▼ ALL BEAR	desk	price above EMA 9, 13, 20, 50, SMA 5, VWAP, previous close, open and PM low; stack 9>13>20>50; **and** the	not aligned

#	Field	Source	Rule	Blank means
			catalyst bullish. Bearish mirror uses the PM high	
11	Catalyst badge	news	⚡ tier 1, ● tier 2, · tier 3, 📈 technical; `+N` more; dimmed when the story is over 4 days old; tooltip: tier, detail, corroborating headlines, age, confidence, evidence	no catalyst in the window: normal

2.3 Signal tags

#	Field	Rule
12	Ten level tags: `9MA` `13MA` `20MA` `50MA` `5MA` `VWAP` `PDC` `OPEN` `PMH` `PML`	▲ green when price is above the level, ▼ red below; tooltip gives the distance in %; no level, no tag
13	Stack	`9>13>20>50` (bull), `9<13<20<50` (bear) or `MA mixed`, with `n/3` in order
14	Month quarter	`Q1 range` … `Q4 range`: the price's position in the 1-month range
15	PM/ADR band	`<0.5`, `0.5-1`, `1-2`, `2-3`, `3+`, with the ratio

2.4 The quote: 32 TradingView columns

Asked for on every scan and every 5-minute refresh:

TradingView column	Card field		TradingView column	Card field
`close`	`price`		`High.1M` / `Low.1M`	`monthHigh` / `monthLow`
`open`	`open`		`High.5D` / `Low.5D`	`weekHigh` / `weekLow` (5 sessions)
`change`	`change` (% today)		`High.3M` / `Low.3M`	`quarterHigh` / `quarterLow`
`change_from_open`	`_changeFromOpen`		`price_52_week_high/low`	`yearHigh` / `yearLow`
`relative_volume_intraday|5`	`rvol` (preferred)		`High.All`	`allTimeHigh`
`relative_volume_10d_calc`	`rvol` when the intraday one is empty or 0		`high` / `low`	`dayHigh` / `dayLow`
`VWAP`	`vwap`		`ATR`	`atr` (14-day)
`EMA9` `EMA13` `EMA20` `EMA50`	`ema9` … `ema50`		`premarket_high/low`	`pmHigh` / `pmLow`
`SMA5`	`sma5`		`market_cap_basic`	`mcap`
`sector` / `industry`	`sector` / `industry`		`float_shares_outstanding`	`floatShares`
`short_percentage_of_float`	never filled by TradingView: see short interest		the row's symbol	`ticker`, `tvSymbol`

Safety checks on every answer: a row whose length differs from the 32 columns asked for refuses the whole batch (D-S3: a shift would move every field by one); a warning when two range windows come back identical for most rows; rows with ATR under 3 % of price are dropped locally, because the scanner cannot compare two columns (a missing ATR is kept).

2.5 Derived numbers and relations

Pure arithmetic on the quote. **A missing input gives a missing output, never a zero** (F-14).

Field	Formula
`prevClose`	`price / (1 + change / 100)`
`gapPct`	`(open − prevClose) / prevClose × 100`
`pmRange`	`pmHigh − pmLow`
`adrPct`	`atr / price × 100`
`pmAdrRatio`	`pmRange / atr` (0 when ATR is 0)
`weekRangePos` … `yearRangePos`	`(price − low) / (high − low) × 100`
`dayRangePos`	`(price − dayLow) / (dayHigh − dayLow) × 100`
`dist<Level>` for EMA 9/13/20/50, SMA 5, VWAP, prev close, open, PM high, PM low	`(price − level) / |level| × 100`, 3 decimals
`vs<Level>`	`above` / `below`
`aboveAllMas` / `belowAllMas`	all four EMAs known, and all above / all below
`maStack`, `maStackScore`	count of EMA9>13, 13>20, 20>50: 3 = `bull`, 0 = `bear`, else `mixed`
`monthQuarter`	month position < 25 `Q1`, < 50 `Q2`, < 75 `Q3`, else `Q4`
`pmAdrBand`	PM/ADR `<0.5`, `0.5-1`, `1-2`, `2-3`, `3+`
Move (× ATR), #94	today’s range ÷ ATR
Days to cover, #20	from FINRA: shares short ÷ average daily volume; from Yahoo (its short ratio) and Nasdaq: as the source reports it

2.6 Volume, float, short interest, borrow

#	Field	Source, in order	Blank means
16	RVOL	TradingView	a fault
17	Market cap	TradingView	a fault
18	Float	TradingView	none for this ticker (new listings)
19	Short % (`shortFloat`), with `shortBasis` (`float` , or `outstanding` when no float: understates it), `shortAsOf` , `shortSrc` , `shortTrend`	**FINRA** file (`cdn.finra.org/equity/regsho/monthly/shrt{date}.txt` , then the OTC biweekly paths; one file for every symbol, cached for the day) → **Yahoo** `quoteSummary` (per symbol; cookie and crumb from `fc.yahoo.com`) → **Nasdaq** `api.nasdaq.com/api/quote/{t}/short-interest` (two weeks of history)	none reported. A success is cached per ticker for the day; a failure is not
20	Days to cover	computed from FINRA's shares short, or as Yahoo / Nasdaq report it	needs short interest (and average volume for FINRA)
—	`shortable` , `easyToBorrow` , `borrowNote`	**Alpaca** assets API, per symbol	null = could not ask; never "no"
—	`regSho` , `regShoDay` , `regShoNote`	**Nasdaq** `nasdaqth{yyyymmdd}.txt` + **NYSE** threshold list, for the settlement day	as above

Borrow answers two different questions that are never merged: *will the broker let you short it now* (Alpaca) and *is it on the Reg SHO threshold list* (a symptom of hard borrow).

Magnitude colours (present / notable / extreme): RVOL ≥ 1.5× / 3× / 5×; float ≤ 50M / 20M / 10M; cap ≤ $2B / $300M / $50M; short ≥ 5 % / 10 % / 20 %; days to cover ≥ 2 / 5 / 10.

2.7 Range

#	Field	Rule	Blank means
21–24	W, M, Q, Y position %, each with its low and high; all-time high as a distance	four bars	Q: under a quarter of history; Y: under a year

Reverse splits. TradingView's longer-window highs are not split-adjusted. When a window's high is more than **10×** the next shorter window's high, that window and every wider one are hidden, and the card says why. The all-time-high distance is suppressed the same way. **Identical windows** (the whole month happened this week) are labelled `= month`.

2.8 News and catalyst

Sources, asked in parallel for every live card:

Source	Endpoint	Needs	Tag
TradingView news	`news-headlines.tradingview.com/v2/headlines`	nothing	TV
Alpaca (Benzinga)	`data.alpaca.markets/v1beta1/news`	key	BZ
Yahoo	`query1.finance.yahoo.com/v1/finance/search`	nothing	YH
SEC EDGAR full-text	`efts.sec.gov/LATEST/search-index`	a User-Agent	SEC
Finnhub	`finnhub.io/api/v1/company-news`	key	FH
Google News RSS	`news.google.com/rss/search`	asked only when the five above returned fewer than 2 items	GN

Filters: stories older than **21 days** are not shown; market round-ups (more than 8 related symbols) are dropped; law-firm class-action notices never count as catalysts; the same headline from several sources is kept once, with its earliest time.

The 39 catalyst categories, matched against each headline of the last **7 days**:

Tier	Categories (sentiment)
1, base 100	Bankruptcy (bear) · FDA Approval (bull) · FDA Rejection (bear) · Trial Win (bull) · Trial Fail (bear) · Earnings Beat (bull) · Earnings Miss (bear) · Guidance Raise (bull) · Guidance Cut (bear) · Takeover Target (bull) · M&A (neutral) · Dilution / offering (bear) · Short Report (bear)
2, base 60	Trading Halt (neutral) · Acquiring (neutral) · Earnings, unclassified (neutral) · FDA Designation (bull) · Upgrade (bull) · Downgrade (bear) · Contract Win (bull) · Partnership (bull) · Buyback (bull) · Reverse Split (bear) · Stock Split (bull) · Index Add (bull) · Index Removal (bear) · Uplisting (bull) · Delisting Risk (bear) · Legal Risk (bear) · Dividend Cut (bear)
3, base 30	Dividend Raise (bull) · Short Squeeze (bull) · Insider Buy (bull) · Insider Sell (bear) · Shelf Filing (bear) · Legal Win (bull) · Product Launch (bull) · Mgmt Change (neutral) · Restructuring (neutral)

Catalyst score = tier base + 8 per corroborating headline (up to 3) + recency (+15 under a day, +5 under two, −10 under four, −25 older). The best is the catalyst; the rest are kept as `others`.

Technical catalysts (the tape, not a story), from the card's own numbers:

Catalyst	Rule
Gap Up / Gap Down	|gap| ≥ 4 % on ≥ 2.5× volume
Big Move	|change| ≥ 2 × ADR % and ≥ 5 %, on ≥ 2× volume
Volume Surge	≥ 5× volume with |change| ≥ 2 %
Monthly Breakout / Breakdown	through the 1-month high or low on ≥ 2× volume
Squeeze Setup	short float ≥ 20 %, up ≥ 3 %, on ≥ 3× volume

A technical catalyst leads only when there is no fresh news, and it never sets bias.

Auto bias: a manual long or short always wins; else a fresh, non-technical tier-1 bull or bear catalyst decides; a tier-2 one decides unless both the short-term market and the sector oppose it; else the market and sector context decide; else "no read". The bias button cycles `⊙ AUTO (long ⚡)` → `▲ LONG` → `▼ SHORT` → AUTO, saved per card; ⚡ means a catalyst decided it.

#	On the card	Blank means
25	Catalyst label	nothing classified: normal
26	Story count	“nothing in three weeks” is a real answer
27	Top headline; the fold lists up to 5 newest with source tag, age and link; warns when a source is not answering; counts the hidden older stories; says when it was fetched	as 26

2.9 Market context and themes

Stamped on every card from the market snapshot of the scan (MARKET.md; legacy MARKET-TAB): `regime`, `regimeLabel`, `longTerm`, `midTerm`, `shortTerm`, `broadResolved` (the card’s TradingView sector mapped to one of the 15 sector ETFs), `secBias`, `secScore`, `secHot`. If SPY and QQQ are both incomplete there is no regime this scan, and the card says so (D-S2).

#	Field	Blank means
89	Regime · long · short, with the sector’s bias and score, and the bias button	a fault

Themes come from two static lists in `src/sideD/themes.js`: a ticker list (NVDA → AI; RKLB → Space; IONQ → Quantum; OKLO → Nuclear; COIN → Crypto; LLY → GLP-1, Pharma; … about 50 tickers) and an industry list (Semiconductors → AI, Semiconductors; Biotechnology → Biotech; Solar Energy → Clean Energy, Solar; … about 25 industries).

2.10 Price, EMAs, ATR, pre-market

#	Field	Source	Blank means
90	Price, previous close, VWAP, open, gap %	TradingView, computed	a fault
91	5-day MA	TradingView	under 5 sessions
92	EMA 9, 13, 20, 50, each green above / red below the price	TradingView	under that many sessions: a new listing has no EMA 50
93	ADR $, ADR %	TradingView, computed	a fault
94	Move (× ATR)	computed	needs ADR
95	Day high / low	TradingView	a fault
96	PM high / low / range	TradingView	no pre-market trading: normal on a thin stock
97	PM / ADR	computed	needs 96 and ADR

2.11 The CANSLIM reading

Fetched by the browser through the tool's `/api/market/oneil/*` routes, which ask qp; also put on the row as 31 `cs*` columns so R1 freezes it. The method behind every value is in CANSLIM; the facts come from the research worker in the new design (§6.5).

The one-line summary (#28–#39): C (EPS %chg, sales %chg, quarter, accelerating, beat +25 %) · A (3-year growth, wobble, ROE) · N (weeks, depth, handle, pivot, % away, checks) · S (U/D, A/D) · L (group, rank, RS in group) · I (funds, change, direction) · M (status, distribution days live, this stock on those days). A missing letter appears on a "Not shown:" line with its reason. **Relative strength:** RS line % off its own high (needs 30 sessions), and divergence (needs 60 sessions and an index low).

C: eight quarters (#40–#49): column header `3 mo to`; EPS $; year-ago EPS (a quarter within 45 days of one year back); %Chg (`n/a (loss a year ago)` is correct: a % from a negative base means nothing); sales; year-ago sales; %Chg (`n/a (no sales a year ago)` for a pre-revenue company); margin (net income ÷ revenue, capped ±999 %); **accelerating** yes/no over the last N quarters; **beat +25 %** X of Y. A missing Q4 is deliberate when EPS and net income disagree in sign (the share count moved).

A: annual (#50–#56): header `12 mo to` (a fiscal year and its own Q4 end on the same day, so the date appears in both tables with different figures); **from 4 quarters** (the year added up from C, `N of 4` when fewer); EPS $; %Chg (`n/a (no prior year filed)` is a gap in filings; `n/a (loss a year ago)` is correct); ROE (`n/a` when equity is zero or negative); 3-year growth (needs 4 unbroken years and a positive base); stability (needs 4 years; low is good); ROE against the **17 %** floor.

N: the weekly base (#57–#64): weeks, depth, waves down, accumulation weeks (0 is an answer), handle ("none yet" is an answer), lip price and week, low price and week, pivot, % away, X of 6 checks. Needs 7 weekly bars.

S: supply (#65–#73): shares outstanding (EDGAR), its basis when not the cover figure, as-of; **share count % change over about a year** with buying back / diluting / flat, always from the same tag (`only one count on file` and `too bunched to give a rate` are answers); float (TradingView) and float as % of shares; short interest with basis and as-of; days to cover; **U/D volume** over N sessions (needs 3, and some down-day volume); **A/D** grade and raw (needs 10). "N sessions, short of 50/65" is the correct label on a young stock.

L: leader or laggard (#74–#79): group name; `rolled up` (ranked at the SIC 3-digit industry group because its 4-digit industry had under 6 ranked names) or `rolled up twice` (the 2-digit major group); rank X of Y; percentile and letter; RS X of N in its group; rotation ▲ into / ▼ out of. Absent for three reasons, only the first fixable by waiting: ranks not built; no RS rating (under a year old: correct); not in the industry map (an ETF, or no SIC).

I: institutions (#80–#83f): distinct **managers** by CIK, unioned across every CUSIP of the issuer and across a 13F-HR and its amendment; change; quarter history; "of N" managers who filed at all that quarter; **share of filers** per quarter; **direction from the share** with a 3 % relative dead band: `rising`, `peaked`, `flat`, `falling`, `turning up` (only `rising` and `turning up` coloured bullish), with the raw holder change printed beside it, uncoloured; newest-quarter %. Warnings when a quarter was counted by filing rather than by manager, or the direction came from the raw count.

M: market direction (#84–#88): status; distribution days live (0 is an answer); this stock on those days, X of Y; verdict **HOLDING UP / GIVING WAY**; a sources line with the date of EDGAR, groups, market and 13F data, each saying "pending" when its file is missing.

2.12 Score and shortlist

The score breakdown (tap the score): the base used (e.g. B4 = 09:40 long), the entry time and bias, the card's regime, which table scored it (a regime `sub` table or the `main` table and why), the bias the table read, its sample count, its confidence, a bar per factor's bucket, and a button to open the table. How it is computed today and why it is replaced: `LEARNING.md` (to write) when written; legacy SCORING-TODAY.

The star (☆/★) adds the card to this tool's shortlist; `★ also T2` shows other tools' stars.

2.13 Display rules

One meaning per colour:

Role	Used for
Direction (green / red)	price against something; up or down on the day. A good score is never green
Magnitude (one hue, 4 intensities)	how big, never whether it is good
Model (orange)	the score, and only the score
Someone else (violet)	another tool’s shortlist
Mine (yellow)	the trader’s star
No claim (grey)	missing data, below a threshold, switched off
Furniture (blue)	links, borders, bar fills; never says anything

Above the cards: search (ticker or sector); screener (all, or one); news (any, has, none, not checked); catalyst (any, has, ⚡ major only, none); aligned (any, fully bullish, fully bearish, either); sort (score, change, RVOL, live first, newest found).

The chart fold mounts a TradingView widget only when the card is on screen and the fold is open; if `tv.js` did not load, it says so (D-S4).

2.14 Per-tool extras

T6 (Unexplained Move) adds `move2hPct` (the move over the last 120 minutes, Alpaca 1-minute bars) and `trend6mPct` (six-month trend before today) from its post-scan tests (TOOLS-AND-SCREENERS.md).

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3. What was planned and never built

The CANSLIM document locked a card specification and built it in phases (CANSLIM §12). Phases 1–5 shipped. What did not:

Item	What it is	Why it matters	In the new design
The ratings row (§7.1): Composite, EPS rating, SMR	the MarketSmith list-view line under the ticker: `Comp 96 · EPS 76 · RS 99 · Grp B+ · SMR B · A/D B+`, plus volume against its 50-day average and `grp 63 of 197 · RS 1 of 13`	the one line most mornings need	built: RS, Grp, A/D, volume vs 50-day and the two group facts exist; Composite, EPS rating and SMR are computed by the research worker from EDGAR
The CANSLIM panel (§11.3)	a full-width surface in letter order (C A N S L I M), each block with its own as-of date, instead of a fold squeezed into a 300–400 px card	an eight-quarter table does not fit in a card column	built as the card's detail view
N, the news algorithm (§10, phase 7)	24 months of company events from EDGAR 8-K items (5.02 management, 8.01/7.01 product, 1.01 contract), openFDA and USASpending; kept only if the price **reacted** (beyond 1 ATR on heavy volume) and the RS line **persisted**; scored with decay 1.0 / 0.9 / 0.7 / 0.4; unconfirmed events listed apart; discarded counts printed	the difference between "what is new about this company" and "what was said this week"	built by the research worker; shown above the existing 3-week headline list, which stays
L extras (L4, L7)	new highs and lows within the group; the sector rank (1 of 33) kept distinct from the group	—	built
S2 float estimate; **I3** funds / banks / management %	no free feed publishes float; Forms 3/4/5 for insiders	—	built, labelled as estimates
Everything explains itself (§13, E1–E8)	every label carries a tappable ⓘ, because `title` tooltips do not appear on a phone; a four-part definition: what it means, how it is calculated, **what it is not**, source and as-of; a Method page	14 written explanations on the card are invisible on Android today	built into every field (§5.6)

4. What failed

ID	Fault	Root cause	Prevented by
—	each tool fetches the same ticker's news, short interest and borrow, and builds its own card	the pipeline lives inside each tool	one shared card (§5.1)
—	while qp restarts, every card's CANSLIM fold is blank, and every scan freezes empty `cs*` columns into R1	O'Neil values computed inside the chart server, asked for on every scan	facts read from the fact store, written by the research worker
—	R1 cannot say when a fact was learned; backtests rebuild CANSLIM "as of" a day	facts carry no time	`known_at` and `as_of` on every fact
D-S1	short interest and borrow wiped on a re-quote	enriched fields not carried over	facts live apart from the quote
D-S2, F-14	missing data read as bearish or as zero	no honest null	`status` on every fact; derived values blank on missing input
D-S3	a column shift could misalign every field	positional parsing	the adapter refuses misaligned batches (kept)
D-S4	the chart fold silently empty when `tv.js` failed	—	says so (kept)
D-D1	no card has a score; the shortlist auto-rule adds nothing	scorer off on every tool	the model applied in-process (§5.3)
D-D2	R4 narrower than R1	a copied table	R4 is a view (`REGISTERS.md` (to write))
D-D6	a one-rung group roll-up shown as an ordinary industry on the Market tab	display	the card's rule used everywhere
D-D7	the short-interest tooltip names the wrong source order	text not generated	tooltips generated from the adapter's order
—	14 field explanations invisible on a phone	`title` attributes	tap ⓘ (§5.6)
—	the industry theme list partly uses Yahoo-style names (`Banks—Regional`, `Drug Manufacturers—General`) that TradingView never sends	a list written against another source	themes keyed on SIC codes and a checked TradingView label list

5. The new design

5.1 Card and tool card

- **The card** is built by the card engine (desk-worker) and is the same for every tool. It is a projection of the fact store: for each fact family, the latest `ok` fact at that instant, or the latest `none` / `unavailable` with its reason.
- **The tool card** holds `tool`, `ticker`, `day`, `screener_keys` (with the version and first-match time of each), `first_seen`, `live`, `star`, `manual_bias`, `score` (with model version, sample size and confidence).
- The page draws a tool’s list from its tool cards, joined to the shared cards.

5.2 Fact families

Each family is fetched, refreshed and judged as one unit. Each is written only by the card engine, except where noted.

Family	Fields	Adapters, in priority	Refreshed	When a source fails
`quote`	the 32 columns (§2.4)	TradingView scanner	at discovery; every 5 min for live and stale cards	the batch is refused; the last quote stays, marked with its age
`derived`	§2.5	computed from `quote`	with every quote	blank where an input is blank
`short_interest`	short shares, `shortFloat`, basis, as-of, trend, days to cover	FINRA → Yahoo → Nasdaq	once a day per ticker; again only after a failure	`unavailable` with each source's reason
`borrow`	shortable, easy to borrow, note	Alpaca	once a day	`unavailable`
`reg_sho`	on the threshold list, settlement day	Nasdaq + NYSE files	once per settlement day (one file for all)	`unavailable`
`news`	items with source, time, link; each source's status	TradingView, Alpaca, Yahoo, EDGAR, Finnhub; Google when fewer than 2	each discovery scan for live cards; at most every 5 min per ticker	the other sources still count; the failing one is named
`catalyst`	catalyst, others, technical catalysts	computed from `news` and `quote`	with `news` and `quote`	—
`bias`	auto bias and its reason	computed from `catalyst` and market context	with either	"no read"
`market_context`	regime, trend stages, sector ETF, sector bias, score, hot	the market engine (§6.4)	each scan	no regime, said plainly
`themes`	theme tags	static lists, keyed on ticker and SIC code	daily	none
`oneil_*`	C, A, N, S, L, I, M, RS, ratings, N events	**the research worker** (§6.5)	nightly; M also on each market-state change	"not built yet · last run DATE"

Family	Fields	Adapters, in priority	Refreshed	When a source fails
`move_tests` (T6)	`move2hPct`, `trend6mPct`	Alpaca bars	at discovery, for the screeners that need them	the screener's test decides (TOOLS-AND-SCREENERS.md)

A fact family that every tool needs is fetched **once per ticker**, whichever tool found it first.

5.3 The card builder

Pure functions, each written once and tested with fixtures:

Function	Input	Output	Spec
`derive(quote)`	quote	derived numbers and relations	§2.5
`rangeSanity(quote)`	quote	hidden windows with reasons; `= month` labels	§2.7
`classifyNews(items)`	news items	catalyst, others	§2.8: the 39 categories, filters, score
`technicalCatalysts(quote, derived, shortInterest)`	—	technical catalysts	§2.8 table
`autoBias(catalyst, marketContext, manualBias)`	—	bias and reason	§2.8 rule
`alignment(derived, catalyst)`	—	ALL BULL / ALL BEAR / none	§2.2
`score(toolModel, card, toolCard)`	the tool's current model	score, sample, confidence, breakdown	`LEARNING.md` (to write)

The score is computed **once, at discovery**, and again only when the trader changes the bias.

5.4 Lifecycle and timing

Event	What happens
a screener matches a ticker for a tool	tool card created or updated (screener key, first match); if the card does not exist today, the card engine builds it, fetching only families that are not fresh
every 5 minutes, 04:00–16:00	one batched TradingView call re-quotes every card of today; `derived` recomputed
each discovery scan	`news` refreshed for live cards; `market_context` re-stamped
a scan in which no screener of the tool matched	the tool card becomes stale
the tool's capture minute	R1 freezes the card and the tool card (`REGISTERS.md` (to write))
after the close	the research worker writes new `oneil_*` facts; tomorrow's cards read them
midnight New York	today's cards and tool cards close; history stays in the database

A restart loses nothing: cards and tool cards are rows, not memory.

5.5 Every value carries its honesty

Each value on a card is one of:

State	Shown as
a value	the value, with its unit and window (`RVOL 6.2×` , `rank 63 of 197` , `U/D 1.41 (50d)`)
`none` : the source answered that there is nothing	the answer in words (`nothing in three weeks` , `no handle yet` , `0 distribution days`)
`unavailable`	grey: `not fetched yet · last tried 09:41 · FINRA timeout`
correct `n/a`	the reason (`n/a (loss a year ago)`)

No blank that reads as zero (CANSLIM X7, E6).

5.6 Presentation

- **The card keeps its layout** (§2.1), its folds, its colour language and its filters. Nothing that is on the card today moves.
- **The ratings row** (§3) sits under the ticker line, always visible.
- **The CANSLIM panel** opens from the row: full width, letter order, each block with its own as-of date, the N event table above the headline list.

- **Every label has a tappable** ⓘ with four parts: what it means, how it is calculated, what it is not, source and as-of. The text is generated from the same definitions the builder uses, so a tooltip cannot disagree with the code (D-D7).
- **A Method page** lists every formula and threshold with its provenance.

5.7 What R1 freezes

The whole card and the whole tool card at the capture minute, plus the version of every screener that matched and the id of the market state used. Every field in §2 is therefore in the photo, and every export is generated from the schema (`REGISTERS.md` (to write)).

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6. Contracts

Owns

Table	Key	Notes
`facts` (the families in §5.2 except `oneil_*` and `market_context`)	subject, family, `known_at`	shared fact store (trunk §5.2)
`cards`	ticker, day	the latest projection, kept for the day
`tool_cards`	tool, ticker, day	per-tool layer

Reads

From	What
tools and screeners	candidates: tool, ticker, screener key and version, match time
market	the current market state
research data	`oneil_*` facts
learning	each tool’s current model

Routes (desk-web)

Route	Returns
`GET /api/tools/:tool/cards?day=`	the tool’s cards for the day, with filters and sort as in §2.13
`GET /api/cards/:ticker?day=`	one card with every family, its status and its as-of
`GET /api/cards/:ticker/canslim?day=`	the CANSLIM panel
`POST /api/tools/:tool/cards/:ticker/star`	star or unstar
`POST /api/tools/:tool/cards/:ticker/bias`	set the manual bias (rescores)

The full API is a leaf of its own (`design/API.md`).

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7. How the rebuild is proven

1. **Field parity.** For a week, the new card engine runs beside one old tool. Each day, every field of every card is compared with the old tool’s R1 photo; every difference is either explained (a fixed fault) or fixed.
2. **The checklist.** Each of the 97 legacy fields has a test: its source, its formula, and its blank reason, from the table in §2.
3. **Fixtures per source.** A recorded answer from each adapter, including a failed one, drives the builder in CI.
4. **Restart test.** Restart desk-worker mid-morning: every card, star, first seen time and score is still there.
5. **qp down test.** Stop quant: every card’s CANSLIM reading is unchanged.